# Remainders modulo 4 for the equation $A(x^2 + 2x^3) = x\,A(x)$

A proof of a conjecture in OEIS A389536

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**Abstract**

There is exactly one power series $A(x) = x + 2x^2 + 2x^3 + 8x^4 + 10x^5 + \cdots$ that satisfies the equation $A(x^2 + 2x^3) = x\,A(x)$. Its coefficients $a(n)$ form the sequence A389536 of the On-Line Encyclopedia of Integer Sequences (OEIS), contributed by Paul D. Hanna. Hanna conjectured that, for $n > 1$, the coefficient $a(n)$ leaves remainder 2 on division by 4 when $n - 1$ is a power of 2, and is divisible by 4 otherwise. We prove this conjecture. The key observation is that, once every coefficient is replaced by its remainder modulo 4, the series $A(x)$ becomes $x + 2x\,C(x)$, where $C(x) = x + x^2 + 2x^3 + 5x^4 + \cdots$ is the generating function of the Catalan numbers. The conjecture then follows from the classical fact that a Catalan number is odd only at positions given by powers of 2. The result has also been checked by computer and proved formally in the Lean proof assistant.

**How this paper was produced.** This paper reports the result of an autonomous AI research effort. ChatGPT (OpenAI), using the GPT-6 model with Pro reasoning and working from prompts by Dev D. Goyal, found the proof, wrote its Lean 4 formalization, and wrote a first draft of the paper. Dev D. Goyal chose the problems and asked for checks and revisions, but did not supply proof ideas. Claude Opus 5.5, used with Claude Code, then checked the work independently: it rebuilt the formal proofs from source and confirmed that they rely on no unproved assumptions, recomputed the sequence from its definition and tested every claim against the published data, and read every step of the proofs. It also rewrote this paper for clarity. Dev D. Goyal has verified the results and is responsible for them. The paper has not been peer reviewed. The section "About this paper" at the end gives the details.

## 1 Introduction

### 1.1 The sequence

The *On-Line Encyclopedia of Integer Sequences* (OEIS) [4] is a large public database of integer sequences, used by mathematicians and scientists to identify sequences and to record what is known about them. Many entries also record *conjectures*: patterns that the contributor observed in the data but could not prove.

This paper is about entry A389536 [1], contributed by Paul D. Hanna in October 2025. The entry reads, in part:

```
G.f.  A(x) satisfies:  A(x) = A(x^2 + 2*x^3) / x.

The radius of convergence r of the g.f.  A(x) is r = 1/2, with A(-r) = 0.

Conjecture:  for n > 1, a(n) == 2 (mod 4) when n = 2^k + 1 (for some k >=
0) otherwise a(n) is divisible by 4.
```

— OEIS A389536, Paul D. Hanna, October 2025

Here "G.f." stands for *generating function*: the terms $a(1), a(2), a(3), \ldots$ of the sequence are stored as the coefficients of the power series $A(x) = a(1)\,x + a(2)\,x^2 + a(3)\,x^3 + \cdots$. In the equation, $A(x^2 + 2x^3)$ means that the polynomial $x^2 + 2x^3$ is substituted for $x$ in $A$ (Section 2 explains this). Multiplying both sides by $x$, we write the equation without division:

$$A(x^2 + 2x^3) = x\,A(x). \tag{1}$$

Equations like this one, which relate $A(x)$ to $A$ evaluated at a higher power of $x$ (here, essentially $x^2$), are called *Mahler-type equations*, after the mathematician Kurt Mahler, who studied them. Equation (1) is *linear*: if $A$ is a solution, so is every multiple of $A$. So one coefficient must be fixed, and the entry fixes $a(1) = 1$. Then all other coefficients are determined (Theorem 3.1). The first terms, starting at $n = 1$, are 1, 2, 2, 8, 10, 12, 32, 80, 202, 356, 540, 944, ...

The second line of the quotation is a statement about $A(x)$ as a function of a real or complex number $x$. This paper treats $A$ only as a formal power series (a list of coefficients), and does not address that statement.

### 1.2 The question

Hanna's conjecture says that, for $n > 1$, the remainder of $a(n)$ on division by 4 is 2 when $n$ is one more than a power of 2 (that is, $n = 2, 3, 5, 9, 17, \ldots$), and 0 otherwise. The first row of the table shows these remainders; the indices $n = 2^k + 1$ are printed in bold. The second row anticipates the main result: it shows the remainders modulo 4 of the coefficients of $x + 2x\,C(x)$, where $C(x)$ is the Catalan series introduced below.

| $n$ | 1 | **2** | **3** | 4 | **5** | 6 | 7 | 8 | **9** | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $a(n)$ mod 4 | 1 | **2** | **2** | 0 | **2** | 0 | 0 | 0 | **2** | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| $[x^n]\,(x+2xC)$ mod 4 | 1 | **2** | **2** | 0 | **2** | 0 | 0 | 0 | **2** | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| $n$ | **17** | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $a(n)$ mod 4 | **2** | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| $[x^n]\,(x+2xC)$ mod 4 | **2** | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| $n$ | **33** | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $a(n)$ mod 4 | **2** | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| $[x^n]\,(x+2xC)$ mod 4 | **2** | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

### 1.3 What this paper proves

The *Catalan numbers* $1, 1, 2, 5, 14, 42, \ldots$ (OEIS A000108) are one of the most common sequences in combinatorics; for example, there are $1, 1, 2, 5, 14, \ldots$ binary trees with $1, 2, 3, 4, 5, \ldots$ leaves. We store them in the series $C(x) = x + x^2 + 2x^3 + 5x^4 + 14x^5 + \cdots$, the only series without constant term that satisfies $C = x + C^2$, and write $c_n$ for its coefficient of $x^n$.

**Main result** (Theorem 3.1). *Modulo* 4*, the series* $A(x)$ *equals* $x + 2x\,C(x)$*. In other words,* $a(n) - 2c_{n-1}$ *is divisible by* 4 *for every* $n > 1$*.*

The number $2c_{n-1}$ leaves remainder 2 on division by 4 when $c_{n-1}$ is odd, and remainder 0 when $c_{n-1}$ is even. A classical fact (Lemma 2.2) says that $c_j$ is odd exactly when $j$ is a power of 2. Together these prove Hanna's conjecture (Corollary 3.2).

### 1.4 What was known

The OEIS entry records the defining equation, an equivalent equation for the compositional inverse of $A$, a formula for $A$ as an infinite product, and numerical values of $A$ as a function. The parity of the Catalan numbers is classical; see for example [2]. We are not aware of an earlier proof of the conjecture.

### 1.5 Why it is interesting

Mahler-type equations have a long history in number theory: Mahler introduced them to prove that certain numbers are transcendental, and they are closely connected with sequences generated by simple machines (automatic sequences) [3]. The equation here is not of the classical form, because of the extra term $2x^3$; but modulo 2 that term disappears, and the equation becomes the classical $A(x^2) = x\,A(x)$. It is a pleasant surprise that the Catalan numbers, which the equation does not mention, describe its solution modulo 4. The method of proof (guess a simple series that solves the equation modulo 4, and use the fact that the solution is unique) is a general one, and it is used throughout this collection of papers.

### 1.6 How to read this paper

Section 2 explains the notions we need: power series, substitution, remainders, the Catalan numbers modulo 2, and a lemma about doubling modulo 4. Section 3 states the results precisely, and Section 4 proves them. Section 5 describes how the results were checked, by computer and by a formal proof in Lean. Section 6 discusses the scope of the result and an open question.

## 2 Background

This section collects the definitions and facts used later. Readers who know formal power series and congruences can skip it and return when needed.

### 2.1 Formal power series

A *formal power series* is an expression

$$U(x) = u_0 + u_1 x + u_2 x^2 + u_3 x^3 + \cdots ,$$

determined by its sequence of coefficients $u_0, u_1, u_2, \ldots$. No number is ever substituted for $x$, so questions of convergence do not arise. We write $[x^n]\,U = u_n$ for the coefficient of $x^n$ in $U$; two series are equal when all their coefficients are equal.

Series are added coefficient by coefficient and multiplied like polynomials:

$$[x^n]\,(UV) = u_0 v_n + u_1 v_{n-1} + \cdots + u_n v_0.$$

Each coefficient of a product is a finite sum. For example, $(1-x)(1+x+x^2+x^3+\cdots) = 1$, so $1 + x + x^2 + \cdots$ is the *multiplicative inverse* of $1 - x$, written $\frac{1}{1-x}$. More generally, every series $D$ with constant term 1 has a unique multiplicative inverse $\frac{1}{D}$, whose coefficients are determined one at a time by $D \cdot \frac{1}{D} = 1$ using only addition, subtraction and multiplication. Every fraction in this paper has a denominator of this kind, so it denotes a power series.

We write $U = O(x^d)$ when all coefficients of $U$ below degree $d$ are zero. So $U - V = O(x^d)$ means that $U$ and $V$ agree up to degree $d-1$. A series of the form $x + u_2x^2 + u_3x^3 + \cdots$, with constant term 0 and coefficient of $x$ equal to 1, is called *normalized*.

### 2.2 Substituting one series into another

If $V(x)$ has constant term 0, we can substitute $V$ for $x$ in $U$:

$$U(V(x)) = u_0 + u_1V(x) + u_2V(x)^2 + u_3V(x)^3 + \cdots .$$

This is called *composition*, also written $U \circ V$. It makes sense because $V(x)^k$ has no terms below $x^k$, so only the finitely many terms with $k \le n$ affect the coefficient of $x^n$. For example, if $U = x + x^2$ and $V = x^2$, then $U(V(x)) = x^2 + x^4$. Composition is associative: $U \circ (V \circ W) = (U \circ V) \circ W$.

Substituting a series into itself repeatedly gives its *iterates*: $U^{\circ 2}(x) = U(U(x))$, $U^{\circ 3}(x) = U(U(U(x)))$, and so on, with $U^{\circ 0}(x) = x$ and $U^{\circ 1} = U$. They must not be confused with *powers* $U^2 = U \cdot U$, $U^3 = U \cdot U \cdot U$. (The OEIS sometimes writes $A^k(x)$ for the $k$-th iterate. In this paper the small circle always means an iterate, and a plain exponent always means a power.)

### 2.3 Remainders and congruences

For an integer $m \ge 2$ we write $a \equiv b \pmod m$ ("$a$ is congruent to $b$ modulo $m$") when $a - b$ is divisible by $m$, that is, when $a$ and $b$ leave the same remainder after division by $m$. For example, $17 \equiv 5 \pmod 4$, since both leave remainder 1, and $-3 \equiv 1 \pmod 4$. The remainder of $a$ on division by $m$ is called its *residue* modulo $m$; it is one of $0, 1, \ldots, m-1$. Saying that $a$ is odd is the same as saying $a \equiv 1 \pmod 2$.

Addition and multiplication are compatible with congruences, so the residues $0, 1, \ldots, m-1$, added and multiplied modulo $m$, form a ring, written $\mathbb{Z}/m\mathbb{Z}$. When $m = p$ is a prime we also write $\mathbb{F}_p$. In $\mathbb{F}_2 = \{0, 1\}$ we have $1 + 1 = 0$, so addition and subtraction coincide.

For power series with integer coefficients, $U \equiv V \pmod m$ means that *every* coefficient of $U - V$ is divisible by $m$. Equivalently, the two series become equal when every coefficient is replaced by its residue; we call this *reducing* the series modulo $m$. The key fact is the following.

*Reduction modulo $m$ is compatible with all the operations above.* To reduce a sum, a product, a substitution, or an inverse, we may reduce the ingredients first and then compute in $\mathbb{Z}/m\mathbb{Z}$. This holds because every coefficient of the result is obtained from finitely many coefficients of the ingredients by adding, subtracting and multiplying, and these operations are compatible with remainders.

**Rings.** Several lemmas below hold over every *commutative ring*, that is, every set with an addition, subtraction and multiplication satisfying the usual rules of arithmetic (for example $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{Z}/m\mathbb{Z}$). No division is used; this matters because in $\mathbb{Z}/m\mathbb{Z}$ division is not always possible (for instance, 2 has no inverse in $\mathbb{Z}/4\mathbb{Z}$).

### 2.4 The method

Almost every proof in this paper follows the same three steps.

1. *Uniqueness.* Show that the defining equation determines the coefficients one at a time, using only addition, subtraction and multiplication. Then the equation has exactly one solution of the required shape, in every commutative ring, including $\mathbb{Z}/4\mathbb{Z}$.
2. *Reduction.* Reducing the integer solution modulo 4 gives a solution of the reduced equation, because reduction is compatible with all operations.
3. *A good guess.* Find a simple series that satisfies the reduced equation and check that it does. By uniqueness, this simple series *is* the reduction of the sequence, and its coefficients can simply be read off.

The creative part is step 3: finding a simple series that works. Step 1 sometimes needs care, when the equation seems to require a division. For a first example, take the Catalan series $C = x + C^2$ from above. Its coefficients $1, 1, 2, 5, 14, 42, \ldots$ are determined one at a time by this equation. Modulo 2, the series $S = x + x^2 + x^4 + x^8 + \cdots$ satisfies the same equation: in $\mathbb{F}_2$, $S^2 = x^2 + x^4 + x^8 + \cdots$ (see Lemma 2.1), so $x + S^2 = S$. Hence $C \equiv S \pmod 2$, and the Catalan coefficient $[x^n]\,C$ is odd exactly when $n$ is a power of 2.

## 2.5 Squaring modulo 2

**Lemma 2.1** (Frobenius identity)**.** *Let $p$ be a prime. For every power series $U$ with coefficients in $\mathbb{F}_p$, we have $U(x)^p = U(x^p)$.*

*Proof.* In $\mathbb{F}_p$, the binomial theorem simplifies to $(a+b)^p = a^p + b^p$, because every binomial coefficient $\binom{p}{i}$ with $0 < i < p$ is divisible by $p$. Also $c^p = c$ for every $c \in \mathbb{F}_p$ (Fermat's little theorem). So $\left(\sum_n u_n x^n\right)^p = \sum_n u_n^p x^{pn} = \sum_n u_n x^{pn} = U(x^p)$. Only finitely many terms matter for any given coefficient, so this is a polynomial identity. □

For $p = 2$ this says that squaring a series modulo 2 doubles every exponent: for instance, $(x + x^3)^2 \equiv x^2 + x^6 \pmod 2$. In particular, the square of any series has, modulo 2, no terms of odd degree.

## 2.6 The Catalan numbers modulo 2

**Lemma 2.2.** *In every commutative ring the equation $U = x + U^2$ has exactly one solution with $U(0) = 0$. Over the integers it is the Catalan series $C(x) = x + x^2 + 2x^3 + 5x^4 + 14x^5 + \cdots$, and*

$$C(x) \equiv x + x^2 + x^4 + x^8 + \cdots = \sum_{k \geq 0} x^{2^k} \pmod 2.$$

*So $[x^n]\,C$ is odd exactly when $n$ is a power of 2.*

*Proof.* The coefficient of $x^n$ in $U^2$ involves only $u_1, \ldots, u_{n-1}$, so the equation determines $u_n$ one at a time: $u_1 = 1$ and $u_n = \sum_{i=1}^{n-1} u_i u_{n-i}$. Over $\mathbb{F}_2$, the series $S = \sum_k x^{2^k}$ satisfies $x + S^2 = x + \sum_k x^{2^{k+1}} = S$ by Lemma 2.1, so by uniqueness $C \equiv S \pmod 2$. □

The parity of the Catalan numbers is classical; see for example [2]. (Here the Catalan number $\frac{1}{n}\binom{2n-2}{n-1}$ appears as the coefficient of $x^n$, so "$n$ a power of 2" corresponds to the familiar statement that the $k$-th Catalan number is odd exactly when $k+1$ is a power of 2.)

### 2.7 Doubling modulo 4

**Lemma 2.3.** *Let $U, V$ be series with coefficients in $\mathbb{Z}/4\mathbb{Z}$. Then $2U = 2V$ if and only if $U$ and $V$ agree modulo 2. Consequently, if $P$ and $Q$ have constant term 0 and $P \equiv Q \pmod 2$, then $2H(P) = 2H(Q)$ in $\mathbb{Z}/4\mathbb{Z}$ for every series $H$.*

*Proof.* For residues $u, v$ modulo 4, $2u = 2v$ means that 4 divides $2(u - v)$, that is, 2 divides $u - v$. Apply this to every coefficient. For the second claim, $H(P)$ and $H(Q)$ agree modulo 2 because reduction is compatible with substitution. □

In other words, *twice a series, modulo 4, depends only on the series modulo 2.* This lets us use information modulo 2 to get information modulo 4, without ever dividing by 2 in $\mathbb{Z}/4\mathbb{Z}$, which is impossible.

## 3 Results

Throughout, $C$ is the Catalan series, the unique normalized series with $C = x + C^2$ (Lemma 2.2), and $c_n = [x^n]\, C$.

**Theorem 3.1.** *(a) Equation (1) has exactly one normalized solution $A$ with integer coefficients (and, more generally, exactly one with coefficients in any commutative ring). The sequence A389536 is $a(n) = [x^n]\, A$.*

*(b) Modulo 4,*

$$A(x) \equiv x + 2x\, C(x) \pmod 4.$$

**Corollary 3.2** (Hanna's conjecture)**.** *For every $n > 1$: if $n = 2^k + 1$ for some $k \geq 0$, then $a(n) \equiv 2 \pmod 4$; otherwise $a(n)$ is divisible by 4.*

*Proof.* For $n > 1$ the coefficient of $x^n$ in $x + 2xC(x)$ is $2c_{n-1}$, so Theorem 3.1(b) gives $a(n) \equiv 2c_{n-1} \pmod 4$. This is 2 modulo 4 if $c_{n-1}$ is odd, and 0 if it is even. By Lemma 2.2, $c_{n-1}$ is odd exactly when $n - 1$ is a power of 2. □

For example, $c_8 = 429$ is odd, and indeed $a(9) = 202 = 4 \cdot 50 + 2$; while $c_9 = 1430$ is even, and $a(10) = 356 = 4 \cdot 89$.

## 4 Proofs

**The idea in brief.** Comparing coefficients in (1) gives a formula for each coefficient in terms of earlier ones, so there is exactly one solution with $a(1) = 1$. For the congruence we guess the answer $D = x + 2xC$ and check that it satisfies (1) modulo 4. Modulo 2 the inner series $x^2 + 2x^3$ is $x^2$, and the Catalan series satisfies $C(x^2) \equiv C(x)^2 \pmod 2$. Because $C$ appears in $D$ multiplied by 2, information modulo 2 is enough (Lemma 2.3), and the identity $C^2 = C - x$ finishes the calculation.

*Proof of Theorem 3.1. (a) Existence and uniqueness.* Write $P(x) = x^2 + 2x^3 = x^2(1 + 2x)$, so that $P^j = x^{2j}(1 + 2x)^j$. By the binomial theorem, the coefficient of $x^{n+1}$ in $P^j$ is

$\binom{j}{n+1-2j}2^{n+1-2j}$ (and 0 unless $2j \le n+1 \le 3j$). So comparing the coefficients of $x^{n+1}$ on the two sides of (1) gives

$$a(n) = \sum_j \binom{j}{n+1-2j} 2^{n+1-2j}\, a(j),$$

where the sum runs over the $j$ with $2j \le n+1 \le 3j$. For $n \ge 2$ all these $j$ satisfy $j \le \frac{n+1}{2} < n$, so this formula computes $a(n)$ from earlier coefficients. For $n = 1$ it reads $a(1) = a(1)$, which is no condition, and for $n = 0$ it gives $a(0) = 0$. Hence, once $a(1) = 1$ is fixed, there is exactly one solution, and it is computed using only additions and multiplications. This works in every commutative ring, and reducing the integer solution modulo 4 gives the solution over $\mathbb{Z}/4\mathbb{Z}$. For example, $a(2) = \binom{1}{1} \cdot 2 \cdot a(1) = 2$, $a(3) = \binom{2}{0} \cdot a(2) = 2$, and $a(4) = \binom{2}{1} \cdot 2 \cdot a(2) = 8$.

*(b) Modulo* 4. By part (a) it suffices to check that the normalized series $D = x + 2x\,C$ satisfies (1) over $\mathbb{Z}/4\mathbb{Z}$. Substituting $P = x^2 + 2x^3$ into $D$ gives

$$D(P) = P + 2P\,C(P) = x^2 + 2x^3 + 2x^2\,C(P) + 4x^3\,C(P) = x^2 + 2x^3 + 2x^2\,C(P) \quad \text{in } \mathbb{Z}/4\mathbb{Z}.$$

To simplify $2\,C(P)$, note that $P \equiv x^2 \pmod 2$. By Lemma 2.3, $2\,C(P) = 2\,C(x^2)$ in $\mathbb{Z}/4\mathbb{Z}$. By Lemma 2.1, $C(x^2) \equiv C(x)^2 \pmod 2$, so again by Lemma 2.3, $2\,C(x^2) = 2\,C^2$. Finally $C^2 = C - x$. Therefore $2x^2\,C(P) = 2x^2(C - x) = 2x^2C - 2x^3$, and

$$D(P) = x^2 + 2x^3 + 2x^2C - 2x^3 = x^2 + 2x^2C = x\,D.$$

So $D$ satisfies (1) over $\mathbb{Z}/4\mathbb{Z}$, and by uniqueness $A \equiv D = x + 2xC \pmod 4$. □

## 5 Checking the results

A mathematical proof can contain mistakes, so the results were checked in two independent ways.

**Numerical check.** We solved (1) coefficient by coefficient with exact integer arithmetic. This reproduces all 34 terms printed in the OEIS entry, and the rule of Corollary 3.2 holds for every $n$ with $1 < n < 130$. The computation was done with the PARI/GP computer algebra system, using code that was written separately from the formal proofs (file `verification/independent_checks.gp` in the accompanying repository [9]). A computation can only test finitely many cases, so it is a check on the definitions and on the indexing, not a proof.

**Formal proof in Lean.** Lean [7] is a *proof assistant*: a program in which mathematical statements and their proofs are written in a precise formal language, and every step is checked mechanically down to the basic rules of logic. All results of this paper are proved in Lean 4 (version 4.19.0) on top of Mathlib [8], Lean's library of formalized mathematics. The statements are collected in the file `lean/Entries/A389536.lean` of the repository, in the namespace `OEIS.A389536`. There the Catalan series appears as `scaledSolution 0`, the solution of a more general equation which, for the parameter value 0, says $C(x - x^2) = x$; the series with this property is the same as the solution of $C = x + C^2$. The proofs are in the module `lean/Families/MahlerParity.lean`. The table lists where each result of this paper is proved.

| Statement in this paper | Lean theorem(s) |
|---|---|
| Theorem 3.1(a) for integer coefficients; $A$ is normalized and satisfies (1) | `mahler_existence`, `mahler_definition`, `mahler_source_unique` |
| Theorem 3.1(b): $A \equiv x + 2xC \pmod 4$ | `mahler_mod_four` |
| Corollary 3.2 | `mahler_residue`, `mahler_four_dvd_iff` |

The Lean project was rebuilt from source for this paper, and Lean was asked to list the basic assumptions (axioms) that each of these theorems ultimately depends on. Every one depends only on the three standard axioms of Lean's logic (`propext`, `Classical.choice`, `Quot.sound`); no step is left unproved. A formal proof certifies exactly the statement that Lean checks. Whether that statement says what the OEIS entry means is a separate question, so the Lean definitions were also compared, side by side, with the definition in the OEIS entry (Section 1.1).

## 6 Discussion

**Scope.** Theorem 3.1 determines the remainder of every coefficient $a(n)$ modulo 4, so it settles the conjecture of the entry completely. The statements in the entry about the radius of convergence and about the values of $A$ as a function are of a different nature (they concern the size of the coefficients, not their remainders) and are not addressed here.

**Modulo 8.** A computer calculation suggests a more intricate rule modulo 8. For every $n$ with $1 < n < 129$ we found: $a(n) \equiv 2 \pmod 8$ when $n - 1$ is a power of 2; $a(n) \equiv 4 \pmod 8$ when $n - 1 = 2^i + 2^j$ with $i \geq j + 2$ (that is, $n = 6, 10, 11, 18, 19, 21, 34, \dots$); and $a(n) \equiv 0 \pmod 8$ otherwise. We have not tried to prove this.

## About this paper

**How the work was done.** This paper is one of 29 papers on conjectures from the OEIS that came out of an autonomous AI research effort directed by Dev D. Goyal [9]. The roles were as follows.

- *Dev D. Goyal* started the project, chose which problems to pursue, asked for proofs, formal verification and revisions, and decided what to publish. Dev D. Goyal did not supply proof strategies or intermediate arguments.
- *ChatGPT* (OpenAI), using the GPT-6 model with Pro reasoning, worked autonomously from those prompts. It searched the OEIS for suitable conjectures, found the proofs, wrote the Lean 4 formalization, and wrote a first draft of this paper.
- *Claude Opus 5.5*, used with Claude Code, checked the work independently and then rewrote this paper. It rebuilt the Lean project from source and confirmed that the theorems listed in Section 5 rely only on Lean's standard axioms; compared the Lean definitions with the OEIS definition; recomputed the sequence with separately written PARI/GP code, compared it with the terms published in the OEIS, and tested every claim of the paper on many terms; and read every step of the proofs. It then rewrote the text to make it self-contained and readable. The theorems and the proof strategies are ChatGPT's. In rewriting, Claude Opus 5.5 expanded the explanations and, in places, wrote out an argument in more detail or by a more elementary route. The Discussion

section was added by Claude Opus 5.5; the further observations in it were checked by computer and are not part of the Lean formalization.

Dev D. Goyal has verified the results and is responsible for this paper.

**Review status.** This paper has not been peer reviewed, and it has not been submitted to a journal or to the OEIS.

**Priority.** An OEIS entry that labels a statement as a conjecture shows only that its contributor did not know a proof; it does not show that no proof existed. During the research only limited searches for earlier work were made. No claim of priority is made.

**Corrections.** Please report any error in the issue tracker of the repository [9], where the Lean sources and the numerical checks are also available.

## Acknowledgements

The sequence, its formulas and the conjecture are due to Paul D. Hanna [1].

## References

[1] P. D. Hanna, entry A389536 (G.f. $A(x)$ satisfies $A(x) = A(x^2 + 2x^3)/x$), *The On-Line Encyclopedia of Integer Sequences*, October 2025. `https://oeis.org/A389536`. Text as of October 2026.

[2] E. Deutsch and B. E. Sagan, Congruences for Catalan and Motzkin numbers and related sequences, *Journal of Number Theory* 117 (2006), 191–215.

[3] J.-P. Allouche and J. Shallit, *Automatic Sequences*, Cambridge University Press, 2003.

[4] N. J. A. Sloane and The OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*. `https://oeis.org`.

[5] R. P. Stanley, *Enumerative Combinatorics*, Volume 2, Cambridge University Press, 1999. (Chapter 5 treats formal power series, composition and compositional inverses.)

[6] I. Niven, Formal power series, *American Mathematical Monthly* 76 (1969), 871–889.

[7] L. de Moura and S. Ullrich, The Lean 4 theorem prover and programming language, in *Automated Deduction – CADE 28*, Lecture Notes in Computer Science 12699, Springer, 2021, 625–635.

[8] The mathlib Community, The Lean mathematical library, in *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020)*, 367–381. Version used: commit `c44e0c8ee63c`.

[9] D. D. Goyal, *Proofs and corrections of OEIS conjectures, with Lean 4 formalizations*, repository, 2026. `https://github.com/dvgyl/oeis-a396843-proof`.